

**Rubric for Performance Indicators of Student Outcome (4):**

**An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts**

| PI #  | Performance Indicator                                                                                   | Beginning                                                                                                                                                                    | Developing                                                                                                                                                                                                  | Proficient                                                                                                                                                                                                           | Exemplary                                                                                                                                                                                                                                          |
|-------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PI-41 | Identify the global, economic, environmental, and societal context of an engineering situation          | <ul style="list-style-type: none"> <li>Unable to identify relevant contexts of the problem.</li> <li>Relevant contexts described in an extremely limited fashion.</li> </ul> | <ul style="list-style-type: none"> <li>One relevant context of the four listed context types identified.</li> <li>The one relevant context described in only a rudimentary fashion.</li> </ul>              | <ul style="list-style-type: none"> <li>Relevant contexts among two or three of the four listed context types recognized.</li> <li>At least two contexts described substantively.</li> </ul>                          | <ul style="list-style-type: none"> <li>Relevant contexts among three or four of the four listed context types identified.</li> <li>At least three of the contexts described thoroughly.</li> </ul>                                                 |
| PI-42 | Describe ethical and professional responsibilities related to an engineering project                    | <ul style="list-style-type: none"> <li>Description of ethical and professional responsibilities absent or extremely limited.</li> </ul>                                      | <ul style="list-style-type: none"> <li>Description of ethical and professional responsibilities is rudimentary.</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>Description of ethical and professional responsibilities is substantive.</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>Description of ethical and professional responsibilities is complete and thorough.</li> </ul>                                                                                                               |
| PI-43 | Explain the impact of engineering decisions in a global, economic, environmental, and societal context. | <ul style="list-style-type: none"> <li>Explanation of relevant impacts of engineering decisions absent or extremely limited.</li> </ul>                                      | <ul style="list-style-type: none"> <li>Explanation of engineering decision impact touches on only one context.</li> <li>Explanation of relevant impacts of engineering decisions is rudimentary.</li> </ul> | <ul style="list-style-type: none"> <li>Explanation of relevant impacts of engineering decisions touches on two to three of the contexts.</li> <li>Explanation is substantive in the majority of contexts.</li> </ul> | <ul style="list-style-type: none"> <li>Explanation of relevant impacts of engineering decisions touches on three or four of the contexts.</li> <li>Explanation is at least substantive in all contexts and is thorough in the majority.</li> </ul> |

**Rubric for Performance Indicators of Student Outcome (5):**

**An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives**

| PI#   | Performance Indicator                                                                                            | Beginning                                                                                                                                                                                                                                     | Developing                                                                                                                                                                                                                                               | Proficient                                                                                                                                                                                                                                                  | Exemplary                                                                                                                                                                                                                                       |
|-------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PI-51 | Establish a collaborative and inclusive environment (Teamwork)                                                   | <ul style="list-style-type: none"> <li>Does not provide encouragement or constructive criticism.</li> <li>Does not listen to other teammates or share knowledge.</li> <li>Does not help other teammates or demonstrate leadership.</li> </ul> | <ul style="list-style-type: none"> <li>Sometimes provides encouragement and constructive criticism.</li> <li>Sometimes listens to other teammates and shares knowledge.</li> <li>Sometimes helps other teammates and demonstrates leadership.</li> </ul> | <ul style="list-style-type: none"> <li>Frequently provides encouragement and constructive criticism.</li> <li>Frequently listens to other teammates and shares knowledge.</li> <li>Frequently helps other teammates and demonstrates leadership.</li> </ul> | <ul style="list-style-type: none"> <li>Always provides encouragement and constructive criticism.</li> <li>Always listens to other teammates and shares knowledge.</li> <li>Always helps other teammates and demonstrates leadership.</li> </ul> |
| PI-52 | Fulfill individual responsibilities and contribute to the team's success. (Individual accountability)            | <ul style="list-style-type: none"> <li>Does not complete individual tasks timely.</li> <li>Does not contribute to the team efforts.</li> <li>Does not interact with other team members.</li> </ul>                                            | <ul style="list-style-type: none"> <li>Completes small number of individual tasks timely.</li> <li>Contributes little to the team efforts.</li> <li>Interacts little with other team members.</li> </ul>                                                 | <ul style="list-style-type: none"> <li>Completes most of the individual tasks timely.</li> <li>Contributes frequently to the team efforts.</li> <li>Interacts regularly with other team members.</li> </ul>                                                 | <ul style="list-style-type: none"> <li>Completes all of the individual tasks timely.</li> <li>Always contributes to the team efforts.</li> <li>Always interacts with other team members.</li> </ul>                                             |
| PI-53 | Define team goals and deadlines, plan tasks, organize & facilitate effective team meetings. (Project management) | <ul style="list-style-type: none"> <li>Does not define any goal or deadline.</li> <li>Does not plan shared or individual tasks.</li> <li>Does not organize nor facilitate any part of any team meeting.</li> </ul>                            | <ul style="list-style-type: none"> <li>Defines at least one goal with a deadline.</li> <li>Plans at least one shared and one individual task.</li> <li>Organizes and facilitates at least one part of one team meeting.</li> </ul>                       | <ul style="list-style-type: none"> <li>Defines a few necessary goals with deadlines.</li> <li>Plans a few necessary shared and individual tasks.</li> <li>Organizes and facilitates a few parts of a few team meetings.</li> </ul>                          | <ul style="list-style-type: none"> <li>Defines several necessary goals with deadlines.</li> <li>Plans several necessary shared and individual tasks.</li> <li>Organizes and facilitates several parts of several team meetings.</li> </ul>      |

**Rubric for Performance Indicators of Student Outcome (6):**

**An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions**

| PI #  | Performance Indicator                                                                                                     | Beginning                                                                                                                                                                                                                                                                                              | Developing                                                                                                                                                                                                                                                                                                                                                                                                   | Proficient                                                                                                                                                                                                                                                                                                                 | Exemplary                                                                                                                                                                                                                                                                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PI-61 | Design an Experiment Plan (How to answer the Driving Question?)                                                           | <ul style="list-style-type: none"> <li>Missing experiment plan</li> <li>Missing driving question</li> <li>Missing identification of key variables</li> <li>Missing data collection procedure.</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>Flawed experiment plan</li> <li>Weak driving question</li> <li>Majority of key variables are not identified</li> <li>Data collection procedure is formulated poorly.</li> </ul>                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Adequate Experiment Plan</li> <li>Driving Question is presented, though it might have minor flaws</li> <li>Almost all variables have been identified</li> <li>Data collection procedure is formulated adequately, but does not account for all externalities.</li> </ul>            | <ul style="list-style-type: none"> <li>Well thought out experiment plan</li> <li>Driving question is appropriately narrow and focused</li> <li>All relevant variables and externalities have been identified</li> <li>Data collection procedure is detailed without being unnecessarily complicated.</li> </ul>                                                                          |
| PI-62 | Acquire data on appropriate variables                                                                                     | <ul style="list-style-type: none"> <li>Data acquisition appears to have significant errors or unrealistic accuracy (fake data?)</li> <li>Data collected for variables that are not part of experiment Plan or some variables are not sampled</li> <li>Missing large portions of data range.</li> </ul> | <ul style="list-style-type: none"> <li>Data acquisition does not include any detail on instrument precision or accuracy performance (sensitivity &amp; calibration)</li> <li>Acquired data is not accompanied by a data acquisition illustration or diagram (test setup not adequately described)</li> <li>Input data range is significantly limited or obviously meaningless for some variables.</li> </ul> | <ul style="list-style-type: none"> <li>Data acquisition includes most instrument capabilities (sensitivity &amp; calibration)</li> <li>Data acquisition setup is illustrated / explained, but a few minor details are missing</li> <li>Input data covers most of the "range of interest" for the key variables.</li> </ul> | <ul style="list-style-type: none"> <li>Data acquisition includes all relevant sensitivity and calibration information</li> <li>Data acquisition setup is carefully and thoroughly explained</li> <li>Input data covers entire range of interest, as well as some additional points / configurations that might be of interest without wasting time on unnecessary procedures.</li> </ul> |
| PI-63 | Interpret experimental data and results with respect to appropriate theoretical models                                    | <ul style="list-style-type: none"> <li>No comparison made, or comparison made to nonsensical models.</li> </ul>                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Weak comparison of data to appropriate model</li> <li>Comparison of data made to model that doesn't include some important relationships among key variables.</li> </ul>                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Adequate comparison made to appropriate model</li> <li>Model includes important relationships among key variables, though some minor details are missing.</li> </ul>                                                                                                                | <ul style="list-style-type: none"> <li>Thorough comparison conducted between sufficiently varied data set and detailed model</li> <li>Theoretical model is sufficiently detailed to provide insight into Driving Question.</li> </ul>                                                                                                                                                    |
| PI-64 | Explain observed differences between model and experiment (bad model, bad measurements, noise, etc.) and draw conclusions | <ul style="list-style-type: none"> <li>Differences are not identified or are incorrectly explained</li> <li>Neither the possibility of using the wrong model nor of collecting erroneous data has been identified</li> <li>Conclusions are not justified.</li> </ul>                                   | <ul style="list-style-type: none"> <li>Most differences are correctly identified, but many are poorly explained</li> <li>Explanation of differences does not consider use of wrong model or possibility of having erroneous data</li> <li>Conclusions are weakly justified.</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li>All major differences are identified; only a few minor differences have been ignored</li> <li>Both model and data have been explored as possible sources of error</li> <li>Conclusions are partially justified by analysis.</li> </ul>                                              | <ul style="list-style-type: none"> <li>All relevant differences have been identified</li> <li>Potential weaknesses in both model and data collection procedure have been identified, but both are well done</li> <li>Conclusions are fully justified by rigorous analysis.</li> </ul>                                                                                                    |

**Rubric for Performance Indicators of Student Outcome (7):****An ability to acquire and apply new knowledge as needed, using appropriate learning strategies**

| PI#   | Performance Indicator                                                          | Beginning                                                                                                                                                                                                                  | Developing                                                                                                                                                                                                                                    | Proficient                                                                                                                                                                                                                                                                         | Exemplary                                                                                                                                                                                                                                                                     |
|-------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PI-71 | Identify necessary techniques, skills and tools for a new situation (research) | <ul style="list-style-type: none"> <li>Identifies a small subset of necessary techniques, skills, and tools</li> <li>Identifies unrelated techniques, skills, and tools.</li> </ul>                                        | <ul style="list-style-type: none"> <li>Identifies some techniques, skills, and tools, but missing some important items</li> <li>Includes some unrelated techniques, skills, and tools.</li> </ul>                                             | <ul style="list-style-type: none"> <li>Identifies almost all of the relevant techniques, skills, and tools</li> <li>Missing some minor techniques, skills, and tools.</li> </ul>                                                                                                   | <ul style="list-style-type: none"> <li>Identifies all relevant techniques, skills, and tools</li> <li>Does not include unrelated techniques, skills, and tools.</li> </ul>                                                                                                    |
| PI-72 | Explain the use of the new techniques, skills and tools (acquisition)          | <ul style="list-style-type: none"> <li>Provides little explanation of how the techniques, skills, and tools should be used</li> <li>Provides incorrect explanation of how to use techniques, skills, and tools.</li> </ul> | <ul style="list-style-type: none"> <li>Explains how some techniques, skills, and tools should be used, but missing some important items</li> <li>Provides some incorrect explanations of how to use techniques, skills, and tools.</li> </ul> | <ul style="list-style-type: none"> <li>Explains how almost all of the techniques, skills, and tools should be used</li> <li>Shows adequate understanding of techniques, skills, and tools</li> <li>Missing the explanation of some minor techniques, skills, and tools.</li> </ul> | <ul style="list-style-type: none"> <li>Explains how all relevant techniques, skills, and tools should be used</li> <li>Shows in-depth understanding of techniques, skills, and tools</li> <li>Does not explain unrelated aspects of techniques, skills, and tools.</li> </ul> |
| PI-73 | Apply the new techniques, skills and tools to the given situation              | <ul style="list-style-type: none"> <li>Applies a small subset of the necessary techniques, skills, and tools</li> <li>Incorrectly applies the techniques, skills, and tools.</li> </ul>                                    | <ul style="list-style-type: none"> <li>Correctly applies some of the techniques, skills, and tools, but missing some important items</li> <li>Incorrectly applies some techniques, skills, and tools.</li> </ul>                              | <ul style="list-style-type: none"> <li>Correctly applies almost all of the techniques, skills, and tools</li> <li>Demonstrates adequate use of techniques, skills, and tools</li> <li>Incorrectly applies some minor techniques, skills, and tools.</li> </ul>                     | <ul style="list-style-type: none"> <li>Correctly applies all relevant techniques, skills, and tools</li> <li>Demonstrates mastery of techniques, skills, and tools</li> <li>Does not apply unnecessary techniques, skills, and tools.</li> </ul>                              |